



Spatiotemporal Assessment of Physicochemical Water Quality in the Al-Aziziyah–Al-Hafariya Water Treatment System, Wasit, Iraq

Muthanna Abdulwahid Ghazi Abd Al Shujairy^{1*}, and Dunia Bahel Gadaan Al-Ghanimy¹

¹ Department of Biology, College of Education, University of Al-Qadisiyah, Iraq

* Corresponding author email: bio.edu.posta24.21@qu.edu.iq

Citation:

Shujairy, M.A.G.A.A.; Al-Ghanimy, D.B.G. Spatiotemporal assessment of physicochemical water quality in the Al-Aziziyah-al-hafariya water treatment system, wasit, Iraq. *ASEAN J. Sci. Tech. Report.* **2026**, 29(10), e265479. <https://doi.org/10.55164/ajstr.v29i10.265479>

Article history:

Received: May 9, 2026

Revised: May 16, 2026

Accepted: May 21, 2026

Available online: September 20, 2026

Publisher's Note:

This article is published and distributed under the terms of the Thaksin University.

Abstract: This study aimed to evaluate the physicochemical characteristics of water within the Al-Aziziyah–Al-Hafariya unified water treatment and distribution system in Wasit Governorate, Iraq. Water samples were collected monthly from six locations representing the raw water source, treatment stages, and the distribution network from September 2025 to February 2026. Key parameters were analyzed, including temperature, turbidity, total suspended solids (TSS), total dissolved solids (TDS), electrical conductivity (EC), salinity, pH, dissolved oxygen (DO), biological oxygen demand (BOD), alkalinity, hardness, residual chlorine, sulfate, nitrate, and phosphate. The results showed significant temporal variations ($P \leq 0.05$) in most parameters, while spatial variations were generally insignificant across all sampling sites. Temperature, TDS, EC, salinity, and hardness were higher during autumn and decreased in winter. In contrast, dissolved oxygen and nutrient concentrations increased during colder months. Turbidity and TSS exhibited seasonal fluctuations. Critically, residual chlorine fell to zero at all six sampling sites during January and February, well below the WHO minimum recommendation of ≥ 0.2 mg/L at the point of delivery, representing a serious public health concern regarding microbiological safety. Statistically significant spatial variation ($P \leq 0.05$) was observed in phosphate concentrations, with elevated levels at distribution-network endpoints (S4–S6, mean 0.238–0.257 mg/L) compared to raw and treated water sites (S1–S2, mean 0.174–0.177 mg/L), suggesting localized contamination within the distribution network. In conclusion, seasonal factors play a dominant role in influencing water quality, while the treatment and distribution system ensures relatively uniform water characteristics across the study area. These findings emphasize the importance of continuous monitoring and optimization of treatment and disinfection processes to ensure safe drinking water quality.

Keywords: Water quality; physicochemical parameters; water treatment plant; distribution system; seasonal variation

1. Introduction

Water is an essential natural resource that plays a vital role in sustaining human health, ecological balance, and economic development. However, water quality deterioration has become a significant global concern due to increasing anthropogenic activities, including urbanization, industrial discharges, and agricultural practices. These activities contribute to the contamination of surface

water resources, particularly in developing countries, and highlight the need for continuous monitoring and evaluation of water quality [1, 2]. Physicochemical parameters such as temperature, turbidity, total suspended solids (TSS), total dissolved solids (TDS), electrical conductivity (EC), pH, dissolved oxygen (DO), biological oxygen demand (BOD), alkalinity, hardness, and nutrients are widely used as key indicators of water quality. These parameters reflect both natural environmental processes and anthropogenic impacts, and they are essential for assessing the performance of water treatment systems and ensuring the safety of drinking water [3-5].

In Iraq, water resources depend mainly on the Tigris and Euphrates rivers, which are increasingly exposed to pollution from untreated wastewater, industrial discharge, and agricultural runoff. Several studies have reported deterioration in water quality across different regions of Iraq. For instance, Farhan et al. [6] assessed the water quality of the Tigris River in Mosul and found variations in physicochemical parameters that affect its suitability for domestic use. Similarly, Alsaadoon et al. [7] evaluated the Diyala River using pollution indices and reported significant changes in water quality due to environmental pressures. Water treatment plants play a crucial role in improving water quality before distribution; however, their efficiency may vary depending on operational conditions and raw water characteristics. Mohammed and Abdulrazzaq [8] evaluated the Al-Wahda water treatment plant in Baghdad and found that some physicochemical parameters were not significantly reduced after treatment. Similarly, Al-Mayah [9] reported that drinking water treatment plants in Kut City exhibited moderate to poor water quality conditions due to increased pollutant loads. In addition, Radhi et al. [10] assessed the same treatment plant and highlighted the importance of using water quality indices to evaluate treatment performance. Furthermore, Hamaamin [11] emphasized that treatment efficiency is strongly influenced by turbidity and disinfection parameters, as well as seasonal variations.

Other studies have demonstrated the impact of environmental factors on treatment plant performance. Abbas et al. [12] showed that increased salinity levels in river water can negatively affect treatment plant efficiency. Similarly, Al-Mohammed et al. [13] applied the Canadian Water Quality Index (CWQI) to evaluate water quality in Karbala and reported variations influenced by both natural and anthropogenic factors. At the distribution level, water quality may change due to infrastructure conditions, environmental factors, and disinfectant decay. Abbas [14] noted that drinking water quality standards in Iraq must be continuously reviewed to ensure compliance with safety requirements, particularly regarding chemical and microbiological parameters. Moreover, residual chlorine stability is a critical factor in maintaining water safety within distribution systems. Dhanabal and Chellaiah [15] reported that chlorine decay is influenced by reactions with organic matter, pipe materials, and water age, which can reduce disinfection efficiency over time. Seasonal variation is considered one of the most important factors influencing water quality. During dry seasons, reduced river discharge and increased evaporation lead to higher dissolved-solids concentrations, whereas wet or colder seasons improve water quality through dilution. These seasonal patterns have been widely observed in Iraqi water systems and are essential for understanding temporal variations in physicochemical parameters [6, 7].

Physicochemical parameters serve as primary indicators of water quality and of the efficiency of treatment processes. However, water quality is a multidimensional concept that also includes microbial aspects. Contamination can occur through multiple pathways, emphasizing the importance of integrated water quality monitoring approaches [16]. Drinking water quality is regulated by international standards that set permissible limits for chemical and microbial contaminants to protect public health [17, 18]. Although prior studies have examined the Al-Aziziyah treatment plant [19] and applied water quality indices to the same facility [10], these investigations focused on treatment-plant-level performance rather than on continuously tracking water quality from raw intake to multiple distribution endpoints. No study to date has combined six-site spatiotemporal sampling spanning the full system—from the Tigris River raw-water intake, through the sedimentation and treatment units, to household taps in the distribution network—within a single seasonal dataset for the Al-Aziziyah–Al-Hafariya project in Wasit Governorate. Relevant regulatory thresholds used for benchmarking in this study include: TDS ≤ 1000 mg/L (WHO guideline: 500 mg/L acceptable); total hardness ≤ 500 mg/L as CaCO₃ (WHO); sulfate ≤ 250 mg/L (WHO); turbidity ≤ 4 NTU at point of delivery

(WHO); residual chlorine ≥ 0.2 mg/L at the consumer tap [17]; and BOD < 2 mg/L for potable water [17]. Iraqi national standards (IQS 417/2009, as revised by Abbas. [14]) largely align with WHO guideline values for these parameters. Therefore, the present study aims to assess the physicochemical characteristics of water within the Al-Aziziyah–Al-Hafariya unified water treatment and distribution system. Specifically, the objectives are: (1) to characterize the spatiotemporal variation of 15 physicochemical parameters across the raw-water intake, treatment stages, and distribution network over six months (September 2025 to February 2026); (2) to compare measured values against WHO and Iraqi national drinking-water standards; (3) to identify critical water-quality risks, particularly concerning residual chlorine and nutrient concentrations; and (4) to provide evidence-based recommendations for optimizing the treatment and disinfection processes.

2. Materials and Methods

2.1 Study Design and Sampling Strategy

This study was designed as a cross-sectional analytical investigation to evaluate water quality across different stages of the Al-Aziziyah–Al-Hafariya water treatment system. Water samples were collected from six sampling sites (S1–S6) representing different stages of the system as follows: S1 represents the raw water intake from the river; S2 and S3 represent treatment stages within the plant (sedimentation and treatment units); while S4, S5, and S6 represent samples collected from the distribution network at different household tap water points outside the treatment plant. Sampling was conducted periodically over six months from September 2025 to February 2026 to capture temporal variation. A total of 108 water samples were collected (6 sampling sites $\times$ 6 months $\times$ 3 replicates), with each sampling point analyzed in triplicate to ensure analytical reliability and reproducibility. Samples were collected using sterile, labeled polyethylene containers following standard sampling procedures to minimize contamination. Immediately after collection, samples were preserved in insulated iceboxes at approximately 4°C and transported to the laboratory within 2–4 hours. All samples were analyzed within 6–12 hours of collection to maintain their physicochemical integrity. The overall experimental workflow of the study, including sampling, laboratory preparation, and physicochemical analysis, is illustrated in Figure 1. The process involved collecting water samples from the Tigris River and the treatment plant, transporting them under controlled conditions, and subsequently analyzing selected parameters in the laboratory.



Figure 1. Experimental workflow of water sampling and physicochemical analysis in the Al-Aziziyah–Al-Hafariya water treatment system.

2.2 Physicochemical Analysis

Water temperature, pH, electrical conductivity ($\mu\text{S}/\text{cm}$ at 25°C), total dissolved solids, salinity, and turbidity were measured in situ with a multi-parameter water quality meter (EZ-9909SP), a pH and turbidity meter (HUAKEYI HK-288), and an EC meter (INESA DDSJ-308F) following APHA Methods 4500- H^+ B, 2510 B, 2540 C, 2520 B, and 2130 B, with salinity and TDS treated as distinct conductivity-derived outputs of the instrument (TDS 531–742.6 mg/L vs. salinity 535.2–750 mg/L in Table 6); total suspended solids were determined gravimetrically on pre-weighed glass-fibre filters dried at $103\text{--}105^\circ\text{C}$ (APHA 2540 D); total hardness was measured by EDTA titration at pH 10 with Eriochrome Black T indicator and calculated as Total Hardness (mg/L as CaCO_3) = $(A \times B \times 1000)/V$, where A is the volume of EDTA used (mL) and B is the mg CaCO_3 equivalent to 1 mL of EDTA (APHA 2340 C); sulfate was determined turbidimetrically at $\sim 420\text{ nm}$ after BaCl_2 addition (APHA 4500- SO_4^{2-} E); phosphate was determined by the ascorbic acid method at $880\text{--}885\text{ nm}$ following Strickland and Parsons [20] and calculated as $\text{PO}_4 = \text{Absorbance} \times F$ (APHA 4500-P E); nitrate was screened semi-quantitatively with AquaChek (HACH) test strips owing to logistical constraints in the remote distribution network, with values ($2\text{--}7\text{ mg/L}$) well below the WHO guideline of 50 mg/L and the Iraqi standard and therefore reported as indicative rather than definitive, with future work to apply APHA 4500- NO_3^- B (220 nm with 275 nm correction); residual chlorine was measured by the DPD colorimetric method (Lamotte kit; APHA 4500-Cl G); dissolved oxygen was determined by Winkler titration on samples fixed immediately with manganese sulfate and alkaline iodide and titrated with standardised sodium thiosulfate (APHA 4500-O); BOD_5 was calculated as the difference between initial and 5-day DO after incubation in the dark at 20°C (APHA 5210 B); and total alkalinity was determined by acid titration with appropriate indicators and expressed as mg/L CaCO_3 (APHA 2320 B).

2.3 Statistical Analysis

Statistical analysis was performed using SPSS software (version 26). Data were expressed as mean $\pm$ standard deviation (SD). Before analysis, data normality was assessed using the Shapiro-Wilk test, and homogeneity of variances was evaluated using Levene's test. ANOVA assumptions (normality and homogeneity of variance) were satisfied for most parameters at $P \leq 0.05$. For BOD, which showed a highly skewed distribution at some sites (e.g., S4–S6 in January), results should be interpreted with caution. One-way analysis of variance (ANOVA) was used to evaluate significant differences among sampling sites and months. The least significant difference (LSD) test was applied to compare means at a significance level of ($P \leq 0.05$).

3. Results and Discussion

3.1 Physical parameters of water

The variation in water temperature across the studied sites and sampling months is presented in Table 1, and the overall treatment-stage profile of selected physical and disinfection parameters along the S1–S6 train is summarised in Figure 2. The results indicated a clear temporal fluctuation, with temperatures ranging from 10.7°C to 34.5°C . The highest temperature was recorded at site S1 during September (34.5°C), while the lowest value was observed at the same site in January (10.7°C). In general, water temperature declined progressively from early autumn into winter, followed by a slight increase in February. The monthly mean values showed significant differences ($P \leq 0.05$), with September and October recording the highest temperatures (27.8°C and 26.6°C , respectively), whereas January had the lowest mean (12.49°C). December and February exhibited intermediate values, with no significant difference between them. In contrast, no significant spatial variation ($P > 0.05$) was observed among the studied sites, as all locations were classified within the same statistical group (a). The overall mean temperature of the study period was $20.6 \pm 8.84^\circ\text{C}$.

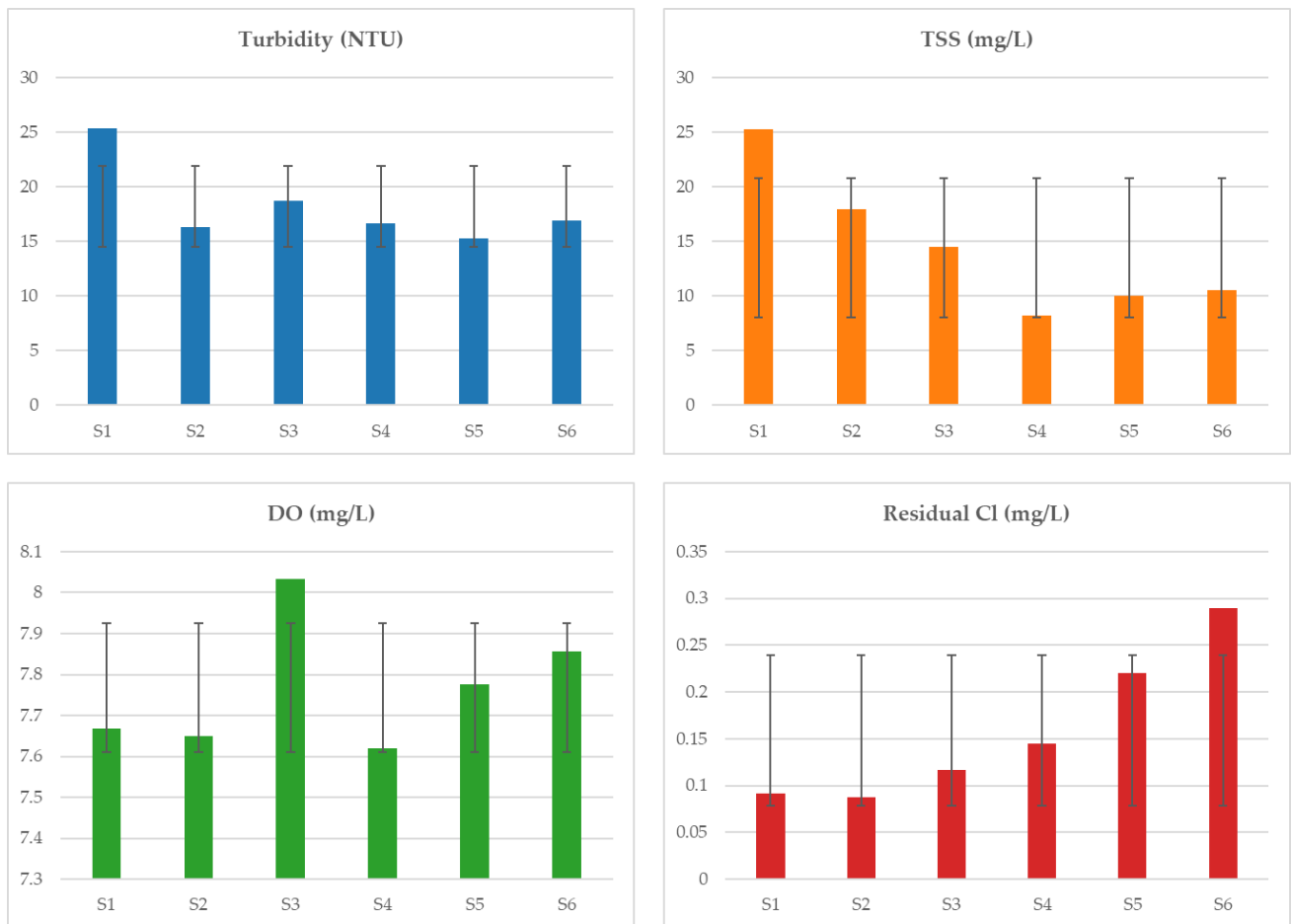


Figure 2. Treatment-stage profile of selected water quality parameters along the Al-Aziziyah–Al-Hafariya treatment train and distribution network (S1: raw water intake; S2: post-sedimentation; S3: post-filtration/treated water; S4–S6: distribution endpoints). Panels show site-wise mean values ($\pm$ SD) of turbidity, total suspended solids, dissolved oxygen, and residual free chlorine, illustrating treatment efficacy and the persistence of turbidity and chlorine depletion along the distribution network.

Table 1. Spatiotemporal variation of water temperature ($^{\circ}\text{C}$) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean $\pm$ SD
S1	34.5	26.8	21.1	16.6	10.7	13.9	20.6 $\pm$ 8.84a
S2	24.6	26.3	21.8	16.6	11.4	14.5	19.2 $\pm$ 5.93a
S3	24.4	25.7	21.0	17.8	11.2	14.4	19.08 $\pm$ 5.68a
S4	27.54	26.98	24.5	17.4	14.16	15.64	21.03 $\pm$ 5.98a
S5	28.3	26.78	24.02	16.5	13.42	15.7	20.78 $\pm$ 6.34a
S6	27.54	27.2	23.32	16.46	14.06	15.68	20.71 $\pm$ 6.05a
Mean $\pm$ SD	27.8 $\pm$ 3.66a	26.6 $\pm$ 0.54a	22.62 $\pm$ 1.52b	16.89 $\pm$ 0.56c	12.49 $\pm$ 1.56d	14.97 $\pm$ 0.79c	
LSD				2.12			7.74

Different letters indicate significant differences at ($P \leq 0.05$).

The turbidity levels measured across the studied sites and sampling months are presented in Table 2. The results revealed noticeable temporal variation, with values ranging from 4.14 to 36.80 NTU. The highest turbidity value was recorded at site S1 in February (36.80 NTU), while the lowest was observed at site S5 in November (4.14 NTU). Generally, turbidity levels were relatively high in September and October, then

decreased sharply in November and January, before increasing again in February. The monthly mean values showed significant differences ($P \leq 0.05$), with October recording the highest turbidity (28.7 NTU), whereas January and November exhibited the lowest (7.46 and 8.75 NTU, respectively). In contrast, no significant spatial differences ($P > 0.05$) were detected among the studied sites, as all locations were grouped within the same statistical category (a). The overall mean turbidity ranged between 15.31 and 25.40 NTU across the sites.

Table 2. Spatiotemporal variation of turbidity (NTU) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean $\pm$ SD
S1	35.50	36.30	13.80	22.50	7.48	36.80	25.40 $\pm$ 12.7 a
S2	24.70	27.50	12.60	6.26	5.11	21.60	16.30 $\pm$ 9.63 a
S3	22.50	26.10	10.60	16.50	6.90	29.60	18.70 $\pm$ 8.91 a
S4	24.00	27.96	6.99	14.82	7.45	18.74	16.66 $\pm$ 8.57 a
S5	23.12	27.80	4.14	12.35	8.67	15.76	15.31 $\pm$ 8.88 a
S6	22.66	26.88	4.34	19.69	9.16	18.78	16.92 $\pm$ 8.50 a
Mean $\pm$ SD	25.4 $\pm$ 5.01a	28.7 $\pm$ 3.75a	8.75 $\pm$ 4.18b	15.35 $\pm$ 5.71c	7.46 $\pm$ 1.42b	23.55 $\pm$ 8.03a	
LSD			6.01				11.39

Different letters indicate significant differences at ($P \leq 0.05$).

Total Suspended Solids (TSS)

The total suspended solids (TSS) recorded across the studied sites and sampling months are presented in Table 3. The results showed clear temporal and spatial variations, with values ranging from 3.2 to 45.2 mg/L. The highest TSS value was observed at site S1 in October (45.2 mg/L), whereas the lowest was recorded at site S5 in February (3.2 mg/L). In general, higher TSS levels were recorded during October and November, followed by a gradual decline during the winter months. The monthly mean values showed significant differences ($P \leq 0.05$), with October exhibiting the highest (19.4 mg/L) and September the lowest (8.64 mg/L). Other months showed moderate values with partial overlap in statistical grouping. Regarding spatial variation, significant differences ($P \leq 0.05$) were observed among sites. Site S1 recorded the highest mean value (25.25 mg/L), while site S4 showed the lowest mean (8.25 mg/L), indicating a reduction in suspended solids along the treatment and distribution system.

Table 3. Spatiotemporal variation of total suspended solids (TSS) (mg/L) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean $\pm$ SD
S1	16.4	45.2	34.4	30.0	12.0	13.5	25.25 $\pm$ 13.3a
S2	13.2	24.8	25.2	22.0	11.5	11.0	17.95 $\pm$ 6.75ab
S3	5.6	16.4	23.2	11.0	11.5	19.5	14.53 $\pm$ 6.39bc
S4	6.56	12.68	4.8	10.56	8.62	6.28	8.25 $\pm$ 2.95c
S5	5.38	8.84	8.92	25.9	7.98	3.2	10.04 $\pm$ 8.08bc
S6	4.72	8.62	9.98	23.5	9.18	7.2	10.53 $\pm$ 6.61bc
Mean $\pm$ SD	8.64 $\pm$ 4.91a	19.4 $\pm$ 13.9b	17.75 $\pm$ 11.5ab	20.49 $\pm$ 7.99b	10.13 $\pm$ 1.73ab	10.11 $\pm$ 5.83ab	
LSD			10.26				9.43

Different letters indicate significant differences at ($P \leq 0.05$).

Total Dissolved Solids (TDS)

The total dissolved solids (TDS) measured across the studied sites and sampling months are presented in Table 4. The results showed relatively consistent spatial values, ranging from 531 to 742.6 mg/L. The highest TDS value was recorded at site S5 in November (742.6 mg/L), while the lowest was observed at site S4 in February (531 mg/L). In general, TDS levels increased during autumn months (October and November), followed by a noticeable decline during winter, particularly in February. The monthly mean values showed significant differences ($P \leq 0.05$), with November recording the highest (738.80 mg/L) and February the lowest (543.97 mg/L). December and January showed intermediate values with partial statistical similarity. In

contrast, no significant spatial differences ($P > 0.05$) were observed among the studied sites, as all locations were grouped within the same statistical category (a). The overall mean TDS values ranged narrowly between 629.67 and 649.83 mg/L across all sites.

Table 4. Spatiotemporal variation of total dissolved solids (TDS) (mg/L) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean $\pm$ SD
S1	649	727	741	576	615	591	649.83 $\pm$ 69.8a
S2	648	721	735	567	572	535	629.67 $\pm$ 84.8a
S3	650	729	738	577	563	534	631.83 $\pm$ 87.5a
S4	648.8	729.6	736.6	581	568.6	531	632.60 $\pm$ 86.6a
S5	648	730	742.6	585.8	561.8	533.6	633.63 $\pm$ 88.1a
S6	648.6	729.8	739.6	589.4	559.6	539.2	634.37 $\pm$ 83.3a
Mean $\pm$ SD	648.7 $\pm$ 0.74a	727.7 $\pm$ 3.47b	738.80 $\pm$ 2.82b	579.37 $\pm$ 7.93c	573.33 $\pm$ 20.9c	543.97 $\pm$ 23.1d	
LSD				15.67			99.2

Different letters indicate significant differences at ($P \leq 0.05$).

Electrical Conductivity (EC)

The electrical conductivity (EC) values recorded across the studied sites and sampling months are presented in Table 5. The results indicated relatively stable spatial values, ranging from 1101.4 to 1561 μ S/cm. The highest EC value was observed at site S3 in October (1561 μ S/cm), while the lowest was recorded at site S4 in February (1101.4 μ S/cm). Generally, EC values were higher during October and November, followed by a gradual decrease toward winter months, reaching minimum levels in February. The monthly mean values showed significant differences ($P \leq 0.05$), with October recording the highest mean (1549.07 μ S/cm), followed by November (1510.37 μ S/cm). In contrast, February exhibited the lowest mean value (1115.2 μ S/cm), indicating a clear seasonal trend. No significant spatial differences ($P > 0.05$) were observed among the studied sites, as all locations were classified within the same statistical group (a). The overall EC values were consistent across all sampling locations, with means ranging between 1311.67 and 1318.67 μ S/cm.

Table 5. Spatiotemporal variation of electrical conductivity (EC) (μ S/cm) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean $\pm$ SD
S1	1347	1530	1515	1178	1202	1140	1318.67 $\pm$ 172a
S2	1338	1553	1491	1201	1167	1120	1311.67 $\pm$ 179a
S3	1340	1561	1512	1204	1166	1114	1316.17 $\pm$ 187a
S4	1334.6	1548.6	1513.4	1208.2	1170.2	1101.4	1312.73 $\pm$ 185a
S5	1336	1550.6	1518.8	1209.8	1163.0	1105	1313.87 $\pm$ 187a
S6	1339.8	1551.2	1512.0	1211.2	1162.8	1111	1314.67 $\pm$ 184a
Mean $\pm$ SD	1339.23 $\pm$ 4.35a	1549.07 $\pm$ 10.2b	1510.37 $\pm$ 9.82c	1202.03 $\pm$ 12.3d	1171.83 $\pm$ 15.03e	1115.2 $\pm$ 13.7f	
LSD				13.53			215.7

Different letters indicate significant differences at ($P \leq 0.05$).

Salinity

The salinity values measured across the studied sites and sampling months are presented in Table 6. The results demonstrated relatively stable spatial values, ranging from 535.2 to 750 mg/L. The highest salinity value was recorded at site S1 during November (750 mg/L), while the lowest value was observed at site S4 during February (535.2 mg/L). In general, salinity levels were elevated during autumn months, particularly in October and November, and then decreased during winter months. The monthly mean values revealed significant differences ($P \leq 0.05$), with November showing the highest mean (740.57 mg/L), followed by October (728.3 mg/L). In contrast, February recorded the lowest mean value (543.00 mg/L). No significant spatial differences ($P > 0.05$) were observed among the studied sites, as all locations were grouped within the

same statistical category (a). The overall mean salinity values were consistent across sites, ranging from 631.83 to 645.67 mg/L.

Table 6. Spatiotemporal variation of salinity (mg/L) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean ± SD
S1	649	728	750	589	594	564	645.67 ± 77.7a
S2	648	725	741	581	563	542	633.33 ± 85.1a
S3	650	713	740	586	562	540	631.83 ± 82.4a
S4	650.4	734.2	740.8	586.6	570.4	535.2	636.27 ± 86.8a
S5	648.4	734.6	738.4	590	558.2	536.4	634.33 ± 87.6a
S6	649.6	735	733.2	592.4	557.6	540.4	634.70 ± 85.5a
Mean ± SD	649.2±0.93a	728.3±8.53b	740.57±5.45c	587.50±3.94d	567.53±13.7e	543.00±10.6f	
LSD				9.87			99.4

Different letters indicate significant differences at ($P \leq 0.05$).

pH

The pH values recorded across the studied sites and sampling months are presented in Table 7. The results indicated slight variations, with values ranging from 7.20 to 8.36, reflecting generally neutral to slightly alkaline conditions. The highest pH value was observed at site S1 in September (8.36), while the lowest was recorded at site S2 in January (7.20). Overall, pH values decreased gradually from early autumn to winter, then increased slightly in February. The monthly mean values revealed significant differences ($P \leq 0.05$), with September exhibiting the highest (8.00) and January the lowest (7.34). Other months showed intermediate values with partial statistical similarity. No significant spatial differences ($P > 0.05$) were observed among the studied sites, as all locations were grouped within the same statistical category (a). The overall pH values remained relatively stable across all sampling points, ranging from 7.57 to 7.69.

Table 7. Spatiotemporal variation of pH values along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean ± SD
S1	8.36	7.85	7.65	7.56	7.31	7.39	7.69 ± 0.38a
S2	8.13	7.95	7.58	7.63	7.20	7.47	7.66 ± 0.33a
S3	8.09	7.84	7.56	7.66	7.37	7.51	7.67 ± 0.25a
S4	7.83	7.66	7.51	7.62	7.36	7.58	7.59 ± 0.15a
S5	7.83	7.72	7.53	7.65	7.42	7.53	7.61 ± 0.14a
S6	7.76	7.61	7.56	7.57	7.40	7.55	7.57 ± 0.11a
Mean ± SD	8± 0.23a	7.77±0.12b	7.56±0.04c	7.62±0.04c	7.34±0.79d	7.50±0.06c	
LSD				0.141			0.298

Different letters indicate significant differences at ($P \leq 0.05$).

Dissolved Oxygen (DO)

The dissolved oxygen (DO) levels measured across the studied sites and sampling months are presented in Table 8. The results showed noticeable temporal variation, with values ranging from 4.6 to 10.1 mg/L. The highest DO value was recorded at site S4 in January (10.1 mg/L), while the lowest was observed at site S1 in November (4.6 mg/L). In general, DO levels decreased during November, followed by a significant increase during the winter months, particularly in January and February. The monthly mean values showed significant differences ($P \leq 0.05$), with January recording the highest mean (9.86 mg/L), followed by February (9.18 mg/L). In contrast, November exhibited the lowest mean value (5.0 mg/L). No significant spatial differences ($P > 0.05$) were observed among the studied sites, as all locations were grouped within the same statistical category (a). The overall mean DO values ranged from 7.62 to 8.03 mg/L across all sites.

Table 8. Spatiotemporal variation of dissolved oxygen (DO) (mg/L) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean ± SD
S1	7.2	7.5	4.6	8.2	9.3	9.2	7.67 ± 1.72a
S2	7.4	6.8	4.9	7.2	10.0	9.6	7.65 ± 1.89a
S3	7.5	8.3	5.2	8.4	9.7	9.1	8.03 ± 1.57a
S4	7.62	7.1	5.14	6.72	10.1	9.04	7.62 ± 1.75a
S5	7.58	7.16	5.06	7.68	10.04	9.14	7.78 ± 1.71a
S6	7.46	7.34	5.10	8.2	10.02	9.02	7.86 ± 1.68a
Mean ± SD	7.46±0.15a	7.36±0.51a	5±0.22b	7.73±0.66a	9.86±0.30c	9.18±0.21c	
LSD			0.460				2.03

Different letters indicate significant differences at ($P \leq 0.05$).

Biological Oxygen Demand (BOD)

The biological oxygen demand (BOD) values measured across the studied sites and sampling months are presented in Table 9. The results showed considerable temporal variation, with values ranging from 0.1 to 6.6 mg/L. The highest BOD value was recorded at site S5 during January (6.6 mg/L), while the lowest values (0.1 mg/L) were observed at multiple sites during September and November. In general, BOD levels were relatively low during autumn months, followed by a noticeable increase during winter, particularly in January. The monthly mean values showed significant differences ($P \leq 0.05$), with January recording the highest (4.49 mg/L) and September the lowest (0.78 mg/L). February and October showed intermediate values with significant variation. In contrast, no significant spatial differences ($P > 0.05$) were observed among the studied sites, as all locations were grouped within the same statistical category (a). The overall mean BOD values ranged from 1.42 to 3.25 mg/L across the studied sites.

Table 9. Spatiotemporal variation of biological oxygen demand (BOD) (mg/L) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean ± SD
S1	1.5	5.8	3.5	3.0	2.9	2.8	3.25 ± 1.41a
S2	2.4	4.0	2.6	1.41	3.2	4.3	2.99 ± 1.07a
S3	0.5	1.9	0.2	0.91	3.1	3.9	1.75 ± 1.49a
S4	0.1	0.62	0.1	0.24	5.74	1.7	1.42 ± 2.20a
S5	0.1	0.82	0.1	0.32	6.6	2.2	1.69 ± 2.53a
S6	0.1	1.12	0.1	0.28	5.4	2.72	1.62 ± 2.10a
Mean ± SD	0.78±0.96a	2.38±2.08ac	1.10±1.53a	1.03±1.07a	4.49±1.61b	2.94±0.99c	
LSD			1.69				2.21

Different letters indicate significant differences at ($P \leq 0.05$).

Total Alkalinity

The total alkalinity values measured across the studied sites and sampling months are presented in Table 10. The results showed moderate temporal variation, with values ranging from 100 to 210 mg/L. The highest alkalinity value was recorded at site S1 in January (210 mg/L), while the lowest was observed at site S3 in September (100 mg/L). In general, alkalinity values increased during winter months, particularly in December and January, compared to autumn. The monthly mean values showed significant differences ($P \leq 0.05$), with December recording the highest mean (197.67 mg/L), followed by January (173.17 mg/L). In contrast, September exhibited the lowest mean value (137.3 mg/L). No significant spatial differences ($P > 0.05$) were observed among the studied sites, as all locations were grouped within the same statistical category (a). The overall mean alkalinity values ranged from 150 to 178.33 mg/L across the sites.

Table 10. Spatiotemporal variation of total alkalinity (mg/L) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean ± SD
S1	180	150	150	200	210	180	178.33 ± 24.8a
S2	180	150	140	200	180	180	171.67 ± 22.2a
S3	100	150	130	200	160	160	150.00 ± 33.4a
S4	102	170	142	200	166	150	155.00 ± 32.7a
S5	144	174	142	206	163	150	163.17 ± 24.2a
S6	118	168	154	180	160	150	155.00 ± 21.04a
Mean ± SD	137.3±36.6a	160.3±11.4b	143±8.36ab	197.67±8.98c	173.17±19.5bc	161.67±14.7b	
LSD	22.68						31.71

Different letters indicate significant differences at ($P \leq 0.05$).

Total Hardness

The total hardness values measured across the studied sites and sampling months are presented in Table 11. The results indicated moderate temporal variation, with values ranging from 472 to 624 mg/L. The highest hardness value was recorded at site S3 in November (624 mg/L), while the lowest was observed at site S1 in February (472 mg/L). In general, hardness values were higher in the autumn months, particularly in October and November, and then decreased during the winter months. The monthly mean values showed significant differences ($P \leq 0.05$), with November recording the highest mean (607.07 mg/L), followed by October (596.67 mg/L). In contrast, February exhibited the lowest mean value (487.67 mg/L), while December also showed a reduced level (500.67 mg/L). No significant spatial differences ($P > 0.05$) were observed among the studied sites, as all locations were grouped within the same statistical category (a). The overall mean hardness values were consistent across sites, ranging from 537.00 to 547.83 mg/L.

Table 11. Spatiotemporal variation of total hardness (mg/L) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean ± SD
S1	532	596	608	528	544	472	546.67 ± 49.6a
S2	531	591	588	484	528	500	537.00 ± 44.2a
S3	533	598	624	524	544	464	547.83 ± 56.8a
S4	533.2	599.4	615	480.2	539.6	501	544.73 ± 53.3a
S5	531.6	597.2	606.6	487	530	488.4	540.13 ± 51.6a
S6	530.4	598.4	600.8	500.8	524.4	500.6	542.57 ± 45.8a
Mean ± SD	531.87±1.10a	596.67±3b	607.07±12.2b	500.67±20.8c	535±8.59a	487.67±16.1c	
LSD	14.68						59.47

Different letters indicate significant differences at ($P \leq 0.05$).

Residual Free Chlorine

The residual free chlorine levels measured across the studied sites and sampling months are presented in Table 12. The results showed low concentrations overall, ranging from 0.00 to 0.94 mg/L. The highest residual chlorine value was recorded at site S6 during September (0.94 mg/L), while a complete absence of chlorine (0.00 mg/L) was observed at all sites during January and February. In general, residual chlorine was detected during the autumn months and gradually declined to zero during winter. The monthly mean values showed significant differences ($P \leq 0.05$), with September recording the highest (0.38 mg/L), followed by December (0.26 mg/L). In contrast, January and February showed zero values, indicating complete depletion of residual chlorine. No significant spatial differences ($P > 0.05$) were observed among the studied sites, as all locations were grouped within the same statistical category (a). The overall mean residual chlorine values ranged from 0.09 to 0.29 mg/L across the sites.

Table 12. Spatiotemporal variation of residual free chlorine (mg/L) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean ± SD
S1	0.05	0.20	0.10	0.20	0.00	0.00	0.09 ± 0.09 a
S2	0.03	0.20	0.10	0.20	0.00	0.00	0.09 ± 0.09 a
S3	0.20	0.10	0.20	0.20	0.00	0.00	0.12 ± 0.09 a
S4	0.27	0.18	0.16	0.26	0.00	0.00	0.15 ± 0.12 a
S5	0.80	0.14	0.16	0.22	0.00	0.00	0.22 ± 0.29 a
S6	0.94	0.14	0.18	0.48	0.00	0.00	0.29 ± 0.39
Mean ± SD	0.38±0.39a	0.16±0.04b	0.15±0.11b	0.26±0a	0±0b	0±0b	
LSD			0.197				0.246

Different letters indicate significant differences at ($P \leq 0.05$).

Sulfate (SO_4^{2-})

The sulfate concentrations measured across the studied sites and sampling months are presented in Table 13. The results showed noticeable temporal variation, with values ranging from 228.12 to 614.75 mg/L. The highest sulfate concentration was recorded at site S2 in December (614.75 mg/L), while the lowest was observed at site S1 in February (228.12 mg/L). In general, sulfate concentrations were relatively high during autumn and early winter months, followed by a marked decrease during January and February. The monthly mean values showed significant differences ($P \leq 0.05$), with December recording the highest mean (469.92 mg/L), followed by November (462.67 mg/L). In contrast, January and February exhibited significantly lower values (259.37 and 278.64 mg/L, respectively). No significant spatial differences ($P > 0.05$) were observed among the studied sites, as all locations were grouped within the same statistical category (a). The overall mean sulfate concentrations ranged from 374.57 to 429.94 mg/L across the studied sites.

Table 13. Spatiotemporal variation of sulfate (SO_4^{2-}) (mg/L) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean ± SD
S1	428.50	448.50	538.12	555.16	267.69	228.12	411.02 ± 136a
S2	444.70	452.90	422.26	614.75	255.20	389.81	429.94 ± 115a
S3	440.20	464.10	426.77	362.31	250.34	303.71	374.57 ± 84a
S4	445.76	449.34	481.25	431.30	260.95	260.11	388.10 ± 100a
S5	445.00	457.49	455.57	412.85	261.33	245.08	379.50 ± 99a
S6	434.88	449.24	452.03	443.15	260.69	245.00	380.80 ± 99a
Mean ± SD	439.8±6.90a	453.6±6.10a	462.67±42.7a	469.92±95.1a	259.37±5.93b	278.64±60.2b	
LSD			58.23				126.3

Different letters indicate significant differences at ($P \leq 0.05$).

Nitrate (NO_3^-)

The nitrate concentrations measured across the studied sites and sampling months are presented in Table 14. The results showed slight temporal variation, with values ranging from 2 to 7 mg/L. The highest nitrate value was recorded at site S1 during October (7 mg/L), while the lowest value (2 mg/L) was observed at sites S3 and S4 during September. In general, nitrate concentrations remained relatively low throughout the study period, with a slight increase during the winter months. The monthly mean values showed significant differences ($P \leq 0.05$), with February recording the highest (4.67 mg/L), followed by January (4.47 mg/L) and December (4.43 mg/L). In contrast, September exhibited the lowest mean value (3.10 mg/L). No significant spatial differences ($P > 0.05$) were observed among the studied sites, as all locations were grouped within the same statistical category (a). The overall mean nitrate values ranged from 3.67 to 4.67 mg/L across the studied sites.

Table 14. Spatiotemporal variation of nitrate (NO_3^-) (mg/L) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean $\pm$ SD
S1	4	7	5	5	4	3	4.67 $\pm$ 1.36a
S2	4	5	4	4	5	5	4.50 $\pm$ 0.54a
S3	2	3	3	4	5	5	3.67 $\pm$ 1.21a
S4	2.6	3.4	2.4	4.8	4.2	5	3.73 $\pm$ 1.10a
S5	3	3.2	2.2	4.8	4.1	5	3.72 $\pm$ 1.09a
S6	3	3.2	2.4	4	4.5	5	3.68 $\pm$ 0.98a
Mean $\pm$ SD	3.10 $\pm$ 0.78a	4.13 $\pm$ 1.58ab	3.17 $\pm$ 1.11ab	4.43 $\pm$ 0.48b	4.47 $\pm$ 0.44b	4.67 $\pm$ 0.81b	
LSD			1.12				1.27

Different letters indicate significant differences at ($P \leq 0.05$).

Phosphate (PO_4^{3-})

The phosphate concentrations measured across the studied sites and sampling months are presented in Table 15. The results showed noticeable temporal and slight spatial variation, with values ranging from 0.013 to 0.533 mg/L. The highest phosphate value was recorded at site S3 in January (0.533 mg/L), while the lowest was observed at site S3 in November (0.013 mg/L). In general, phosphate levels were low during autumn months, followed by a significant increase during winter, particularly in January. The monthly mean values revealed significant differences ($P \leq 0.05$), with January recording the highest (0.42 mg/L) and November the lowest (0.03 mg/L). Other months exhibited intermediate values with varying statistical groupings. Unlike most other parameters, slight significant spatial differences ($P \leq 0.05$) were observed among some sites, with S4 and S5 recording relatively higher mean values than S1 and S2. The overall mean phosphate values ranged from 0.174 to 0.257 mg/L across the studied sites.

Table 15. Spatiotemporal variation of phosphate (PO_4^{3-}) (mg/L) along the treatment stages and distribution system of Al-Aziziyah–Al-Hafariya water project

Sites	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026	Mean $\pm$ SD
S1	0.096	0.266	0.024	0.214	0.244	0.217	0.177 $\pm$ 0.09 a
S2	0.250	0.183	0.021	0.119	0.302	0.171	0.174 $\pm$ 0.09 a
S3	0.318	0.186	0.013	0.177	0.533	0.137	0.227 $\pm$ 0.17 b
S4	0.317	0.209	0.049	0.158	0.520	0.292	0.257 $\pm$ 0.16 c
S5	0.318	0.231	0.022	0.217	0.456	0.187	0.238 $\pm$ 0.14 b
S6	0.273	0.217	0.026	0.240	0.448	0.175	0.230 $\pm$ 0.13 b
Mean $\pm$ SD	0.26 $\pm$ 0.08a	0.22 $\pm$ 0.03ac	0.03 $\pm$ 0.01b	0.19 $\pm$ 0.04c	0.42 $\pm$ 0.11d	0.20 $\pm$ 0.05c	
LSD			0.079				0.164

Different letters indicate significant differences at ($P \leq 0.05$).

3.2 Discussion

The results of the present study demonstrated that physicochemical parameters are primarily influenced by temporal variation rather than by spatial differences across the treatment and distribution systems, as visualized in the seasonal heatmap of all measured parameters in Figure 3. This indicates that environmental and seasonal factors play a dominant role in determining water quality, while treatment processes help maintain relatively uniform conditions across the studied sites. The absence of significant spatial variation in parameters such as turbidity, total dissolved solids (TDS), electrical conductivity (EC), salinity, and hardness suggests that the treatment system operates consistently across different stages. However, when evaluated against WHO [17] and Iraqi national standards [14], several parameters frequently exceeded permissible thresholds. TDS values ranged from 543 to 739 mg/L, with many monthly means exceeding the WHO aesthetic guideline of 500 mg/L, though all remained below the absolute maximum of 1000 mg/L. Total hardness was consistently elevated (472–624 mg/L as CaCO_3), frequently surpassing the WHO guideline of 500 mg/L, with all sites exceeding this threshold in October and November. Sulfate concentrations were markedly elevated during autumn–early winter (mean up to 469.92 mg/L in December), far exceeding the WHO limit of 250 mg/L; sulfate levels approached acceptable levels only in January and

February. Turbidity at site S1 (raw water intake) reached 36.80 NTU in February and 35.50 NTU in September, and even at downstream distribution points, mean site-level turbidities (15.31–25.40 NTU) far exceeded the WHO recommendation of ≤ 4 NTU at the point of delivery. The persistence of elevated turbidity throughout the distribution network—without progressive improvement from S1 (raw water) to S6 (household tap)—suggests that the conventional treatment is insufficient for turbidity removal under certain seasonal conditions, posing risks for disinfection efficacy and consumer health. These threshold exceedances are important public health concerns that require urgent attention from water utility management. Similar findings were reported in Iraqi water treatment plants, where physicochemical characteristics showed limited variation between treatment units and distribution points, indicating stable operational performance [19, 21, 22]. However, this apparent stability does not necessarily indicate high removal efficiency, as conventional treatment processes are generally more effective in removing suspended particles than dissolved constituents. The persistence of dissolved parameters, such as TDS and EC, throughout the treatment and distribution system observed in this study is consistent with findings from several recent studies. Mesfun and Derib [23] and Mendez-Ruiz et al. [24] indicated that dissolved ions are not significantly reduced during conventional treatment processes, as their removal requires advanced treatment technologies. Similarly, Nasier and Abdulrazzaq [25] demonstrated that while turbidity removal efficiency is high in conventional treatment plants, dissolved solids remain relatively unchanged. The significant temporal variation observed in most physicochemical parameters reflects the influence of environmental and seasonal conditions. The increase in TDS, EC, salinity, and hardness during warmer periods can be attributed to increased evaporation and reduced dilution. Similar seasonal patterns have been reported in Iraqi water systems, where water quality parameters fluctuate due to climatic conditions and hydrological changes [26,27]. These findings are also supported by global studies that highlight the role of environmental factors in influencing water quality variability [28].

Parameter	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan 2026	Feb 2026
Temperature (°C)	27.8	26.6	22.6	16.9	12.5	15.0
Turbidity (NTU)	25.4	28.8	8.7	15.4	7.5	23.5
TSS (mg/L)	8.6	19.4	17.8	20.5	10.1	10.1
TDS (mg/L)	649	728	739	579	573	544
EC (µS/cm)	1339	1549	1510	1202	1172	1115
Salinity (mg/L)	649	728	741	588	568	543
pH	8.00	7.77	7.57	7.62	7.34	7.51
DO (mg/L)	7.46	7.37	5.00	7.73	9.86	9.18
BOD (mg/L)	0.78	2.38	1.10	1.03	4.49	2.94
Total Alkalinity (mg/L CaCO ₃)	137	160	143	198	173	162
Total Hardness (mg/L CaCO ₃)	532	597	607	501	535	488
Residual Cl (mg/L)	0.382	0.160	0.150	0.260	0.000	0.000
Sulfate (mg/L)	440	454	463	470	259	279
Nitrate (mg/L)	3.10	4.13	3.17	4.43	4.47	4.67
Phosphate (mg/L)	0.262	0.215	0.026	0.188	0.417	0.197

Color intensity is normalized per row (min–max) so each parameter uses its own scale; cell labels show raw site-averaged means.

Figure 3. Seasonal heatmap of physicochemical water quality parameters across the sampling months (September–February) at the Al-Aziziyah–Al-Hafariya water treatment system. Cell shading shows the monthly mean of each parameter (averaged across sites S1–S6), with warm colors indicating higher concentrations or values and cool colors indicating lower values. The heatmap highlights the dominance of temporal over spatial variation and the contrast between warm-season (high turbidity, sulfate, TDS, hardness) and cold-season (depleted residual chlorine, elevated BOD at distribution endpoints, higher DO) water-quality signatures.

In contrast, dissolved oxygen (DO) levels were higher during colder months, likely due to increased oxygen solubility at lower temperatures. This behavior is consistent with the findings of Bwapwa et al. [29], who reported that temperature significantly influences oxygen dynamics in water systems. The inverse relationship between temperature and DO observed in the present study is well established and reflects physicochemical principles. The variation in biological oxygen demand (BOD) observed in this study may be related to seasonal changes in organic matter and microbial activity. Increased BOD values may indicate higher organic load or enhanced microbial processes within the water system. Notably, BOD values at distribution endpoints S4–S6 during January (5.74, 6.6, and 5.4 mg/L, respectively) substantially exceeded the WHO drinking-water guideline of < 2 mg/L, suggesting possible organic loading or microbial activity within the distribution network during winter months. This finding warrants further investigation, including microbiological analysis, to assess potential contamination risks. Similar observations were reported by Jumaa and Abdulqader [30], who found that environmental conditions and water quality characteristics in Iraqi urban systems influence BOD levels. Residual chlorine represents the most critical public health concern identified in this study. Complete depletion of residual chlorine (0.00 mg/L at all six sites) during January and February constitutes a serious violation of the WHO guideline of ≥ 0.2 mg/L at the point of delivery [17] and the Iraqi national drinking-water standards [14]. The public health implications are significant: the absence of residual chlorine provides no barrier against microbial recontamination during distribution, potentially exposing consumers to waterborne pathogens. Several factors may account for this wintertime depletion: (a) insufficient chlorine dosing at the treatment plant during colder months when operators may underestimate chlorine demand; (b) increased chlorine demand driven by elevated organic matter and turbidity loads observed in November–February; (c) extended water residence time within the distribution network under conditions of reduced consumption; and (d) lower reaction kinetics at lower temperatures that nonetheless allow greater net consumption at prolonged contact times. In comparison, Dhanabal and Chellaiah [15] highlighted that maintaining chlorine stability remains a significant challenge in distribution networks under varying environmental conditions, emphasizing the importance of adaptive dosing strategies. Furthermore, Elsherif et al. [31] demonstrated that hydraulic conditions and network complexity significantly affect disinfectant distribution. Immediate remediation recommendations include: increasing chlorine dosing during the November–February period; installing online residual chlorine monitors at key distribution nodes; and conducting seasonal chlorine demand profiling to adjust dosing schedules proactively. The statistically significant spatial variation in phosphate concentrations ($P \leq 0.05$) is noteworthy, as phosphate was the only parameter to exhibit such spatial differentiation. Mean phosphate concentrations at distribution endpoints S4 (0.257 mg/L), S5 (0.238 mg/L), and S6 (0.230 mg/L) were appreciably higher than at the raw-water site S1 (0.177 mg/L) and treated-water site S2 (0.174 mg/L). This progressive elevation along the distribution pathway—rather than at the raw-water intake—strongly suggests post-treatment contamination, most plausibly from pipe corrosion inhibitors (many of which are phosphate-based), from leaching of aging infrastructure, or from infiltration of phosphate-laden groundwater through pipe defects. Dedicated investigation of the distribution pipes at S4–S6, including infrastructure inspection and source-specific phosphate isotope analysis, is recommended to confirm the contamination route. Similar findings of localized contamination of distribution networks have been reported in Iraqi and regional studies, where localized variations in water quality were associated with infrastructure conditions and environmental factors [22,32]. Several studies have emphasized that the efficiency of water treatment plants is influenced by both raw water quality and operational conditions. Al-Mayah [19] reported that treatment efficiency in the Al-Aziziyah plant is affected by variations in influent water quality. Similarly, Choudhury and Saha [33] highlighted that the operational efficiency of treatment plants depends on multiple factors, including design, maintenance, and environmental conditions. These findings are further supported by recent studies that emphasize the importance of continuous monitoring and optimization of treatment processes [34]. In addition, broader environmental studies have indicated increasing pressure on water resources in Iraq due to pollution and climate change. Ethaib et al. [35] reported that water resources in southern Iraq face significant challenges, including contamination and reduced availability. Similarly, Ali and Abojassim [36] highlighted the presence of heavy metals in drinking water sources, indicating potential health risks. Overall, this study's results

demonstrate that seasonal environmental factors primarily control water quality in the studied system, while treatment and distribution processes help maintain spatial consistency. However, the observed depletion of residual chlorine and localized variation in certain parameters highlight critical issues that require improved monitoring and management to ensure a safe drinking water supply.

4. Conclusions

The results of this study indicate that the physicochemical properties of water in the Al-Aziziyah–Al-Hafariya project are mainly influenced by seasonal changes rather than spatial differences. Most parameters showed higher values during autumn and decreased in winter, while dissolved oxygen and nutrient levels increased during colder months. The absence of significant spatial variation in most parameters reflects the efficiency of the treatment processes and the stability of the distribution system. However, the complete depletion of residual chlorine during winter suggests potential limitations in disinfection efficiency. Additionally, the observed variation in phosphate concentrations indicates possible localized influences within the distribution network. Comparison with the WHO [17] and Iraqi national (IQS 417/2009) standards revealed recurring exceedances of sulfate, total hardness, TDS, and turbidity—parameters not adequately addressed by conventional treatment alone. Overall, the water quality is relatively stable across the system, with seasonal factors being the primary driver of variation. Based on the findings, the following specific recommendations are made: (1) chlorine dosing should be increased during November–February to maintain residual chlorine ≥ 0.2 mg/L at all distribution endpoints; (2) online residual chlorine monitors should be installed at key nodes in the distribution network to enable real-time detection of depletion events; (3) infrastructure at distribution endpoints S4–S6 should be inspected for pipe defects and potential phosphate contamination sources; (4) coagulant dosing and filtration processes should be reviewed to consistently achieve WHO turbidity targets (≤ 4 NTU) at point of delivery; and (5) advanced treatment steps (e.g., nanofiltration or ion exchange) should be evaluated for reducing sulfate and hardness to within WHO guidelines. The study has several limitations: it was conducted at a single treatment plant over 6 months; nitrate was measured using semi-quantitative test strips with limited precision; no GPS coordinates or site map were included; and microbiological parameters were not assessed. Future research should incorporate microbiological monitoring, extend the study period to capture interannual variability, apply a Water Quality Index (WQI) framework for integrated assessment, and include a spatial map of sampling sites to improve reproducibility and verification.

5. Acknowledgments

It is with great pleasure that the author expresses his deep gratitude to the supervisor from the Department of Biology, College of Education, University of Al-Qadisiyah, for his guidance and constant support throughout the research process. Thanks are also offered to the personnel at the water purification station who cooperated with the author in collecting samples.

Author Contributions: Conceptualization, M.A.G.A. and D.B.G.A.; methodology, M.A.G.A. and D.B.G.A.; validation, M.A.G.A. and D.B.G.A.; formal analysis, M.A.G.A.; investigation, M.A.G.A.; resources, M.A.G.A. and D.B.G.A.; data curation, M.A.G.A.; writing—original draft preparation, M.A.G.A.; writing—review and editing, M.A.G.A. and D.B.G.A.; visualization, M.A.G.A.; supervision, D.B.G.A.; project administration, D.B.G.A. Both authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Conflicts of Interest: The authors declare no conflict of interest.

References

- [1] Cain, N. L.; Gleick, P. H. The Global Water Crisis. *Issues Sci. Technol.* **2005**, *21*, 79–82.
- [2] Gleick, P. H.; Cooley, H. Freshwater Scarcity. *Annu. Rev. Environ. Resour.* **2021**, *46*, 319–348. <https://doi.org/10.1146/annurev-environ-012220-101319>

- [3] American Public Health Association (APHA). *Standard Methods for the Examination of Water and Wastewater*, 23rd ed.; American Public Health Association: Washington, DC, 2017
- [4] Boyd, C. E. *Water Quality: An Introduction*; Springer: New York, 2000. <https://doi.org/10.1007/978-1-4615-4485-2>
- [5] Al Yousif, M. A.; Chabuk, A. Assessment Water Quality Indices of Surface Water for Drinking and Irrigation Applications—A Comparison Review. *J. Ecol. Eng.* **2023**, *24*, 40–55. <https://doi.org/10.12911/22998993/161194>
- [6] Farhan, A. F.; Al-Ahmady, K. K.; Al-Masry, N. A. A. Assessment of Tigris River Water Quality in Mosul for Drinking and Domestic Use by Applying CCME Water Quality Index. *IOP Conf. Ser.: Mater. Sci. Eng.* **2020**, *737*, 012204. <https://doi.org/10.1088/1757-899X/737/1/012204>
- [7] Alsaadoon, D. W. K.; Hassan, F. M.; Mahdi, W. M. Assessment of Water Quality of Diyala River Using Overall Index of Pollution (OIP) in Iraq. *Iraqi J. Agric. Sci.* **2023**, *54*, 682–690. <https://doi.org/10.36103/ijas.v54i3.1748>
- [8] Mohammed, N. Y.; Abdulrazzaq, K. A. Evaluation of Drinking Water Quality in Al Wahda Treatment Plant in Baghdad City—Iraq. *J. Eng.* **2021**, *27*, 38–50. <https://doi.org/10.31026/j.eng.2021.09.04>
- [9] Al-Mayah, W. T. Chemical and Microbial Health Risk Assessment of Drinking Water Treatment Plants in Kut City, Iraq. *Mater. Today: Proc.* **2021**, *42*, 3062–3067. <https://doi.org/10.1016/j.matpr.2020.12.920>
- [10] Radhi, I.; Abduljaleel, H.; Mohmmmed, Z.; Abdulhassan, A.; Jazaa, B. Assessment of Al-Wahda Water Treatment Plant within Baghdad City. *AIP Conf. Proc.* **2024**, 020070. <https://doi.org/10.1063/5.0236870>
- [11] Hamaamin, Y. A. Performance Evaluation of Conventional Water Treatment Plants Using Turbidity and Disinfection Parameters. *Zanco J. Pure Appl. Sci.* **2026**, *38*, 177–188. <https://doi.org/10.21271/ZJPAS.38.1.14>
- [12] Abbas, S. H.; Khudair, B. H.; Jaafar, M. S. River Water Salinity Impact on Drinking Water Treatment Plant Performance Using Artificial Neural Network. *J. Eng.* **2019**, *25*, 149–159. <https://doi.org/10.31026/j.eng.2019.08.10>
- [13] Al-Mohammed, F. M.; Al-Saadi, R. J. M.; Al-Fawzy, A. M.; Mohammed-Ali, S. H.; Mutasher, A. A.; Hommadi, A. H. The Analysis of Water Quality Using Canadian Water Quality Index: Green Belt Project/Kerbala-Iraq. *Int. J. Des. Nat. Ecodyn.* **2021**, *16*, 91–98. <https://doi.org/10.18280/ijdne.160112>
- [14] Abbas, H. A. Revisited Iraqi Drinking Water Quality Standards (IQS 417/2009). *Tikrit J. Eng. Sci.* **2025**, *32*, 23. <https://doi.org/10.25130/tjes.32.1.23>
- [15] Dhanabal, G.; Chellaiah, G. Enhancing Residual Chlorine Stability in Water Distribution Systems: Challenges, Modeling Approaches, and Advanced Optimization Strategies. *Next Chem. Eng.* **2025**, *1*, 100005. <https://doi.org/10.1016/j.nxcen.2025.100005>
- [16] Alegbeleye, O. O.; Singleton, I.; Sant'Ana, A. S. Sources and Contamination Routes of Microbial Pathogens to Fresh Produce during Field Cultivation: A Review. *Food Microbiol.* **2018**, *73*, 177–208. <https://doi.org/10.1016/j.fm.2018.01.003>
- [17] World Health Organization. *Guidelines for Drinking-Water Quality*, 4th ed.; World Health Organization: Geneva, 2017.
- [18] U.S. Environmental Protection Agency. *Drinking Water Regulations and Contaminants*; U.S. Environmental Protection Agency: Washington, DC, 2020.
- [19] Al-Mayah, W. T. Evaluation of Efficiency of Drinking Water Treatment Plant in Al-Aziziyah City, Iraq. *Sci. Res. J. Agric. Life Sci.* **2022**, *2*, 1–7. <https://doi.org/10.47310/srjals.2022.v02i02.001>
- [20] Strickland, J. D. H.; Parsons, T. R. *A Practical Handbook of Seawater Analysis*; Fisheries Research Board of Canada: Ottawa, 1972.
- [21] Khader, B.; Owaied, Y. H.; Jalal, R. Physical and Chemical Analysis of Water Quality at Al-Dur Water Treatment Plant. *J. Primeasia* **2024**, *5*, 1–9. <https://doi.org/10.25163/primeasia.519766>
- [22] Hamza, A.; Hama, B. H.; Karim, M. M.; Ahmed, K. Water Quality Assessment for Qaladze Water Treatment Plant, Sulaymaniyah Governorate, Kurdistan Region, Iraq. *Adv. Civ. Eng. Environ. Sci.* **2024**, *1*, 15–24.
- [23] Mesfun, N. M.; Derib, S. D. Performance Evaluation and Factors Influencing a Drinking Water Treatment Plant, Shire Indassilassie, Ethiopia. *Water-Energy Nexus* **2024**, *7*, 235–242. <https://doi.org/10.1016/j.wen.2024.08.001>

- [24] Mendez-Ruiz, J. I.; Barcia-Carreño, M. B.; Mejía-Bustamante, L. J.; Cornejo-Pozo, Á. K.; Salas-Vázquez, C. A.; Valverde-Armas, P. E. Assessment of the Performance of a Water Treatment Plant in Ecuador: Hydraulic Resizing of the Treatment Units. *Sustainability* **2023**, *15*, 1235. <https://doi.org/10.3390/su15021235>
- [25] Nasier, M.; Abdulrazzaq, K. A. Performance Evaluation of the Turbidity Removal Efficiency of Al-Muthana Water Treatment Plant. *J. Eng.* **2022**, *28*, 1–13. <https://doi.org/10.31026/j.eng.2022.03.01>
- [26] Hamaamin, Y. A.; Abdullah, J. B. Temporal Variation of Drinking Water Quality Parameters for Sulaimani City, Kurdistan Region, Iraq. *UHD J. Sci. Technol.* **2020**, *4*, 99–106. <https://doi.org/10.21928/uhdjst.v4n2y2020.pp99-106>
- [27] Hussein, M. A.; Muhsun, S. S.; Abudi, Z. N. Groundwater Quality Assessment for the Wells System in Zurbatiyah, Iraq, for Civil and Irrigation Uses by Two Water Quality Index Approaches. *J. Ecol. Eng.* **2025**, *26*, 389–402. <https://doi.org/10.12911/22998993/207193>
- [28] Kumar, P. Water Quality Assessments for Urban Water Environment. *Water* **2021**, *13*, 1686. <https://doi.org/10.3390/w13121686>
- [29] Bwapwa, J. K.; Mkhize, N.; Seyam, M. Evaluation of Operational Efficiency and Performance for a Water Treatment Plant. *S. Afr. J. Chem. Eng.* **2024**, *49*, 11–34. <https://doi.org/10.1016/j.sajce.2024.04.003>
- [30] Jumaa, M. W.; Abdulqader, R. S. Evaluation of Drinking Water Quality Index in Kirkuk City, Iraq. *Cent. Asian J. Med. Nat. Sci.* **2024**, *5*, 815–823.
- [31] Elsherif, S. M.; Kazma, M. H.; Taha, A. F. Quality-Aware Hydraulic Control in Drinking Water Networks via Controllability Proxies. *arXiv* **2024**, arXiv:2401.12214
- [32] Abdul-Kareem, N.; Al-Baidhani, J. H. Assessing the Influence of Climate Change on Water Treatment Efficiency in Baghdad, Iraq. *J. Wildl. Biodivers.* **2023**, *7*, 713–729
- [33] Choudhury, S.; Saha, A. K. Prediction of Operation Efficiency of Water Treatment Plant with the Help of Multi-Criteria Decision-Making. *Water Conserv. Sci. Eng.* **2018**, *3*, 79–90. <https://doi.org/10.1007/s41101-017-0034-2>
- [34] Silva, G. M. E.; Campos, D. F.; Brasil, J. A. T.; Tremblay, M.; Menciondo, E. M.; Ghiglieno, F. Advances in Technological Research for Online and In Situ Water Quality Monitoring—A Review. *Sustainability* **2022**, *14*, 5059. <https://doi.org/10.3390/su14095059>
- [35] Ethaib, S.; Fahs, M.; Mishbak, H.; Fares, M. N.; Makki, J. S.; Alhello, A.; Bitar, A. A. Water Resources in South of Iraq: Current State, Future Evolutions, Challenges, and Potential Solutions. *Hydrology* **2026**, *13*, 87. <https://doi.org/10.3390/hydrology13030087>
- [36] Ali, B. H.; Abojassim, A. A. Heavy Metals in Healthy Water Samples Used by Newborn in Al-Najaf Governorate, Iraq. *Al-Bahir J. Eng. Pure Sci.* **2025**, *6*, 7. <https://doi.org/10.55810/2313-0083.1097>